Re-Engineering the Student Renewal → Class Scheduling Pipeline

AI Program Manager (Operations) · take-home case for Bambinos.live · Deliverables A, B, D (report) + C (working prototype, in the GitHub repo)

1 of 12
paid renewals reached a scheduled class by the next morning

₹1,26,850
of paid renewals waiting, oldest 67 h

3 of 12
paid invoices never reached a valid teacher form

8 of 12
would need zero human touch in the redesigned flow

Executive summary

I analysed the 13 invoices, 12 teacher forms, 4 booking requests and 2 schedule rows in the provided dataset (1-3 Nov 2026, evaluated at the Head of Ops' 4 Nov 09:00 morning view). **The 4-table chain does not leak because people are slow. It leaks because it asks people to re-type data the system already has, with no owner and no clock between steps.** Teachers who do file the form do it fast (median 27 min after payment); Ops approves fast when it looks (median 1.5 h). The damage comes from what falls *between* those steps.

What I found

- **The teacher form is the main leak.** 2 paid invoices have no form at all (84906, 84910), 1 form cites an invoice that does not exist (digits swapped), 1 was submitted twice. Teachers are bimodal: fast, or never.
- **Ops approval is a pull queue with no timer.** 8 form rows sit pending, all >12 h, oldest 47.5 h. Approved ones took 0.75-2.5 h, so capacity is not the issue; visibility is.
- **Two failures are system logic, not human error.** A same-batch renewal failed with 'batch full' because the batch is full *of the student renewing* (5505). A batch-change renewal has no owner at all (5512).
- **Money authority is informal.** A 20% discount with no approval trail, a half-paid installment recorded by the teacher as fully paid, and a 'paid' form backed only by a UPI screenshot while the invoice says Failed.

What I recommend

- **Eliminate Step 2 and Step 3 for happy-path renewals.** Capture renewal intent when the teacher generates the payment link (batch, plan, discount code); let the verified payment webhook trigger a rules engine that creates the booking and schedule rows and confirms to parent and teacher by WhatsApp.
- **Route every exception to exactly one named owner with an SLA timer**, using 14 reason codes, with the evidence attached so the owner decides in one click.
- **Ship value in week 1 without any automation**: a daily 'paid-not-scheduled' list for the Head of Ops, and a fix to the capacity check.

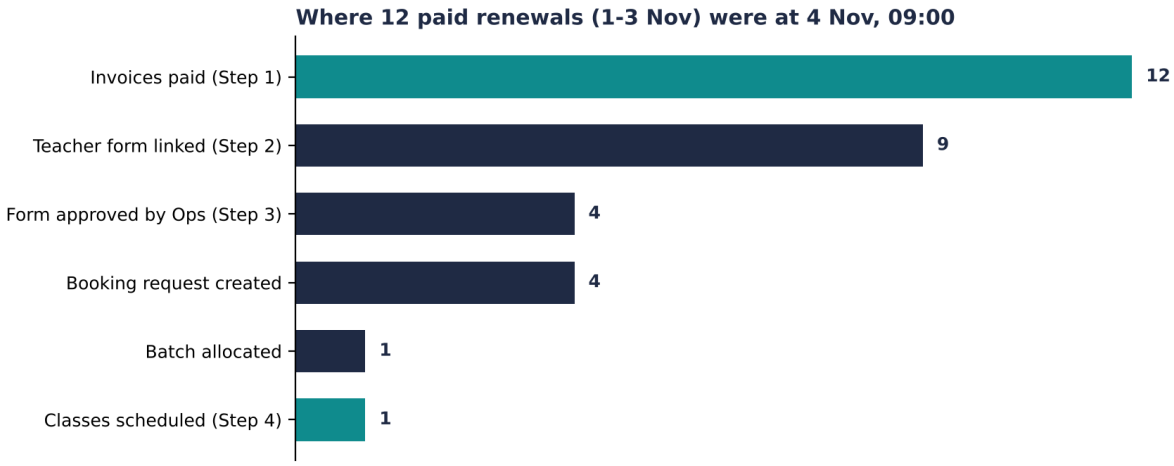
What I built

- A working reconciliation engine (Python, stdlib only) that outputs auto-approved rows, flagged exceptions with reason codes, and 23 drafted WhatsApp/Slack messages; 13/13 decisions verified by a golden test. Result: **5 auto-approved, 3 one-tap parent confirmations, 4 human exceptions, 1 already complete.**

Caveat I want to be upfront about: 13 invoices over 3 days is a sample, not a population. Every rate in this report is indicative; Day 1-30 of the plan replaces them with 8-12 weeks of production data.

Deliverable A - Root-cause problem definition & leakage audit

Method: join every invoice to its form, booking request and schedule rows by ID (and fuzzy-match when IDs fail), then ask at each hop 'what is the first thing that can go wrong here, and who would notice?'. Evidence below is from the dataset, not hypothetical.



A1 · Hidden failure modes found in the data

#	Failure mode	Evidence in the data	Root cause	Impact on parent
1	**Orphaned invoice**	84906 (₹18,000, 41 h) and 84910 (₹11,000, 22 h): paid, no form in the table	Process depends on a teacher remembering; no trigger or timer	Paid, silence, class gap
2	**Wrong invoice reference**	Form 3313 cites 84931; the real invoice is 84913 (digits swapped). The pasted link is wrong too	Free-text re-keying of an ID	Lookup fails, Ops must chase
3	**Duplicate form**	3308 and 3309, same invoice, 5 min apart ('browser froze')	No idempotency on submit	Risk of double booking
4	**Unclocked approval queue**	8 form rows pending, oldest 47.5 h; approved ones took 0.75-2.5 h	Ops pulls from a sheet; no SLA or alert	Class not scheduled
5	**False 'batch full'**	Booking 5505: batch 9205 is 6/6, and student 68105 *is* one of those 6	Capacity check counts the renewing student as a new seat	Paid, then told 'full'
6	**Stuck allocation**	Booking 5502 'Pending_Batch_Allocation' for 64 h; same batch, 5/6 seats; zero judgement needed	A human step that needs no human	Class gap
7	**Batch change has no owner**	5512: classbatch_id NULL, 'Awaiting_Parent_Batch_Selection'. No Gita batch at Sat/Sun 11:00 exists in the reference data	Nobody owns slot offering or capacity	Waits indefinitely
8	**Payer ≠ registered parent**	84904: invoice phone …77, other name and email vs registered …04; only the teacher's remark explains it	Payment identity is never reconciled	Delay or wrong account
9	**Discount without authority trail**	84907: 20% (₹1,900) 'retention scholarship'; no approval field in any table. Form shows net ₹7,600, so gross-vs-form checks false-alarm	Discounts are verbal, not coded	Revenue leakage / disputes
10	**Partial payment recorded as full**	84909: installment 1 of 2, ₹5,250 paid of ₹10,500; form says ₹10,500 and 24 sessions	Teacher records the plan, not the receipt	Full term on half payment
11	**'Paid' without payment truth**	84911: invoice Failed, no gateway id; teacher filed a form 10 min later citing a UPI screenshot	Screenshot accepted as proof	Money in limbo or unpaid
12	**Data hygiene**	Cyrillic look-alike letters in 84913's parent name; form id 3306 missing from the sequence; Stripe invoices for UAE/US parents stored in INR	No validation at write time	Silent match failures

Not visible in a 3-day sample but certain at hundreds of renewals a week: *weekend and night approvals (the sample spans Sun-Tue; 2 of the sampled batches run Sat/Sun); teacher on leave or left; parent pays twice or asks for a refund after approval; siblings under one parent account; mid-term plan upgrades; GCC/US batch times (the schema has no time zone);*

term-end timing so that a gap opens before the renewed classes start. Also worth asking: invoice 84901 shows only 2 schedule rows for 24 purchased sessions - is scheduling rolling, or incomplete even on the 'good' path?

A2 · Four systemic root causes

- **Re-keying:** the form re-enters student, class, amount and invoice that already sit in *invoices*. Every re-key is a typo, a duplicate or a no-show waiting to happen (modes 1-3).
- **No owner, no clock:** between each table there is a hand-off, and nothing measures how long a record sits there (modes 4, 6, 7).
- **Business rules live in people's heads:** discounts, partial plans, payer identity and 'who counts toward capacity' are applied informally (modes 5, 8-11).
- **Intent is captured after payment, not before:** whether the child continues in the same batch is only learned when the form arrives, so nothing can be automated until a human has typed it.

A3 · Quantifying the risk: metrics and SLAs (broken or invisible today)

Metric	Definition	Sample reading (4 Nov 09:00)	Proposed SLA / target	Owner
Payment-to-Schedule TAT	paid_at → first class_schedule_bookings row	1 of 12 done (1.75 h). 11 open, oldest 66.7 h	P90 ≤ 4 h; zero-touch ≤ 15 min	Head of Ops
Renewal Completion Rate	% paid renewals scheduled at the check time	8% (1/12)	≥ 95% within 24 h	Head of Ops
Orphaned Invoice Rate	Paid renewals with no valid linked form after 2 h / paid renewals	25% (3/12: 2 never filed, 1 wrong id)	0% once form is removed; ≤ 2% during transition	Ops + Faculty lead
Form-to-Approval Age	Form created → approved; count and age of pending	Median 1.5 h; 8 pending, all >12 h, oldest 47.5 h	Clean forms auto in minutes; manual ≤ 2 h	Ops lead
Stuck Invoices >12 h	Paid, not scheduled, older than 12 h	11 of 12 paid	0 without a named owner and timer	Ops lead
Revenue Waiting	Σ net_amount of paid, not scheduled	₹1,26,850 (11 invoices)	< 5% of weekly renewal value at 9:00	Head of Ops
Touch Rate	% renewals needing any manual action	100% today (teacher form + Ops approval for every one)	≤ 30% (sample projects ~33%)	Head of Ops
Bad / duplicate form rate	(duplicate + wrong-id forms) / forms	17% (2/12)	0% (form retired)	Faculty lead
False Capacity Failure	Booking failed 'full' while student already in batch	25% (1/4 booking requests)	0%	Product / Eng
Exception SLA breach	Open exceptions past reason-code SLA	Not measurable today (no owner, no clock)	≤ 5%	Ops lead
Class Interruption Days	First renewed class date − last old-term class date	Not in the data; instrument in Day 1-30	≤ 0 days for same-batch	Head of Ops

Tiers by revenue exposure: the highest-value stuck items in the sample are GCC/US renewals (84906 ₹18,000, 84908 ₹22,000), which are likely the hardest for Ops to chase across time zones (to confirm with Ops) - one reason to make confirmation self-serve by WhatsApp.

Deliverable B - AI & no-code re-engineered workflow ('zero-touch')

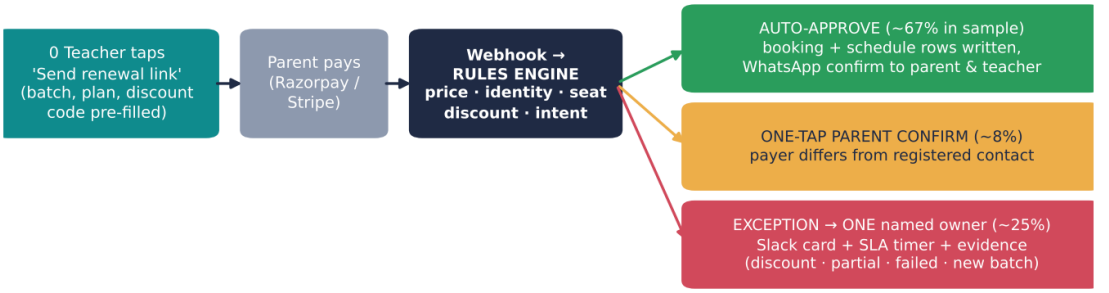
B1 · Question each step before automating it

Step	Why it exists today	Verdict	Redesign
1 Invoice	Parent pays via link	**Keep, enrich**	The teacher generates the link from a 'Send renewal link' button that pre-fills student, batch, plan and price and carries **intent** (same batch / change request) and an optional **discount code**. Prevention beats detection: typos, wrong amounts and unauthorised discounts cannot be created.
2 Teacher form	Invoice lacks intent; Ops does not trust payment data	**Eliminate** (happy path)	Intent moves to link creation. For legacy flows the form stays optional and pre-filled, then retires after Wave 3. Teachers get an instant 'renewal confirmed, credited to you' message, which removes their reason to skip it.
3 Ops verification	Check 'right student, right amount, really paid'	**Automate**	These are deterministic joins on a signature-verified webhook. A rules engine does them in seconds; Ops only sees exceptions.
4 Batch & schedule	Someone must pick a batch	**Automate / self-serve**	Same batch: automatic (net new seats = 0). Batch change: a WhatsApp list message of 2-3 Ops-curated slots; Ops intervenes only when no slot exists.

AS-IS: 3 manual hand-offs, no clock, no owner



TO-BE: intent captured BEFORE payment, rules engine decides, humans handle exceptions only



Every decision logs rule_version, evidence and timestamps → event log → Metabase dashboard. Kill-switch reverts to manual.

Shadow mode first: the engine writes to staging only until it matches Ops decisions for 2 weeks.

B2 · Zero-touch criteria (all must be true)

Payment captured and signature verified · gross amount equals list price and discount (if any) matches a valid, approved code · invoice student/user/educator/class IDs match master data · same batch and same slot · payer identity matches or is confirmed · no other open invoice for the student in the last 7 days · renewing student's seat retained. In the sample this is **8 of 12 paid renewals (67%)**: 84901, 84902, 84903, 84905, 84906, 84908, 84910, 84913. One more (84904) needs a one-tap payer confirmation; three (84907 discount, 84909 partial, 84912 batch change) need a human decision.

B3 · Smart exception handling: detect, triage, alert exactly one owner

Each reason code carries a detection rule, an automatic first action, one accountable owner and an SLA. The engine attaches the evidence string, so the owner never has to open four tables.

Reason code	Detection rule	Automatic first action	Owner → escalation	SLA
PAYER_CONTACT_MISMATCH	Invoice phone/email/name ≠ registered record, IDs match	WhatsApp the **registered** number: 'reply YES'	Parent → Ops agent	4 h
ORPHANED_INVOICE	Paid, no form linked (id or fuzzy) after 2 h	Parent one-tap 'same batch?' + teacher nudge	Parent → Ops agent → Ops lead	4 h
DISCOUNT_NO_APPROVAL_REF	discount_amount > 0 with no valid code	Slack card to Ops lead; teacher asked for approver	Ops lead / Finance	4 h
PARTIAL_PAYMENT	payment_status = Partially_Paid	Hold full-term booking; draft pro-rata option; parent instalment link	Ops lead / Finance	4 h
FORM_AMOUNT_MISMATCH	Form amount ≠ invoice net_amount	Teacher told to correct; engine trusts invoice	Teacher → Ops	4 h
PAYMENT_NOT_CONFIRMED	Invoice not Paid but form says paid	Poll gateway order; if captured → auto-approve, else fresh link to parent	Payments → Ops lead	4 h
BATCH_CHANGE_REQUESTED	continue_same_batch = No, or slot ≠ current	Offer 2-3 slots by WhatsApp; Slack card if none exist	Ops scheduling	6 h
PRICE_MISMATCH / ID_MISMATCH	Gross ≠ list price; IDs disagree with master	Block, alert; never auto-fix money or identity	Ops lead	4 h
Auto-fixed, logged only: INVOICE_ID_TYPO, DUPLICATE_FORM, FORM_APPROVAL_BACKLOG, STUCK_PENDING_ALLOCATION, FALSE_BATCH_FULL	Secondary-key match; earliest-wins; clean form past 2 h; same-batch pending; renewing student already holds the seat	Fix and write evidence to the audit log	Engine (random 5% daily audit by Ops)	0

B4 · Where AI is used, and where it is deliberately not

- **Deterministic rules decide money, identity and capacity.** An LLM cannot approve a payment, which makes the system auditable and cheap.
- **LLM #1 - remark reader:** turns free text ('Mother paid from her number', 'UPI processing screenshot', Hinglish) into structured signals that lower or raise risk. Prompt: prompts/triage_prompt.md.
- **LLM #2 - message drafter:** adapts tone and language (English / Hindi / Hinglish) around a locked template; facts are injected, never generated. Prompt: prompts/message_drafting_prompt.md.

B5 · Guardrails that stop automation from making things worse

- **Shadow mode first** (writes to staging, compared daily against Ops decisions); the bot can only create rows, never edit or delete; every write stamps rule_version and evidence.
- **Idempotency** on gateway_payment_id (no double bookings); **kill-switch** flag reverts to manual; auto-pause if a wrong-schedule or double-booking event occurs.
- **Message hygiene:** max 2 parent messages per invoice per day, quiet hours 21:00-08:00 for parents, send to the *registered* number for identity checks, amounts verified against the invoice before send.

Tooling: n8n (self-hosted) or Make for orchestration, MySQL read/write via a service account, WhatsApp Business API templates, Slack, Metabase on the read replica. Node-by-node mapping: docs/n8n_workflow.md.

Deliverable C - Working prototype: results on the provided dataset

Repo: **github.com/<your-username>/bambinos-renewal-reconciliation** · run **python src/reconcile.py** · 13/13 golden test passing. The engine starts from each *invoice* so it also catches the ones with no form.

Invoice	₹ net	Engine decision	Reason codes	What happens next
84901	12,000	**COMPLETE**	-	No action; end-to-end in 1.75 h
84902	10,500	**AUTO-APPROVE**	STUCK_PENDING_ALLOCATION	Allocate to current batch; schedule; WhatsApp confirm
84903	9,000	**AUTO-APPROVE**	FORM_APPROVAL_BACKLOG	Auto-approve clean form; schedule; confirm
84904	12,000	**PARENT CONFIRM**	PAYER_CONTACT_MISMATCH, FORM_APPROVAL_BACKLOG	WhatsApp registered number; Slack only if no YES in 4 h
84905	11,000	**AUTO-APPROVE**	FALSE_BATCH_FULL	Override false 'full' (net new seats 0); schedule; confirm
84906	18,000	**PARENT CONFIRM**	ORPHANED_INVOICE	Parent one-tap + teacher nudge; Slack to Ops at 4 h
84907	7,600	**EXCEPTION**	DISCOUNT_NO_APPROVAL_REF, FORM_APPROVAL_BACKLOG	Slack Ops lead: approve/reject 20% discount; ask teacher for approver
84908	22,000	**AUTO-APPROVE**	DUPLICATE_FORM, FORM_APPROVAL_BACKLOG	Dedupe 3309; auto-approve 3308; schedule; confirm
84909	5,250	**EXCEPTION**	PARTIAL_PAYMENT, FORM_AMOUNT_MISMATCH, FORM_APPROVAL_BACKLOG	Hold full term; Ops decides pro-rata; teacher corrects form; parent instalment link
84910	11,000	**PARENT CONFIRM**	ORPHANED_INVOICE	Parent one-tap + teacher nudge; Slack at 4 h
84911	12,000	**EXCEPTION**	PAYMENT_NOT_CONFIRMED, FORM_APPROVAL_BACKLOG	Poll gateway; fresh link to parent; teacher told to hold
84912	9,000	**EXCEPTION**	BATCH_CHANGE_REQUESTED	Offer slots to parent; Slack Ops scheduling (no matching batch in data)
84913	11,500	**AUTO-APPROVE**	INVOICE_ID_TYPO, FORM_APPROVAL_BACKLOG, homoglyph (info)	Re-link 3313 → 84913 (all keys match); auto-approve; schedule

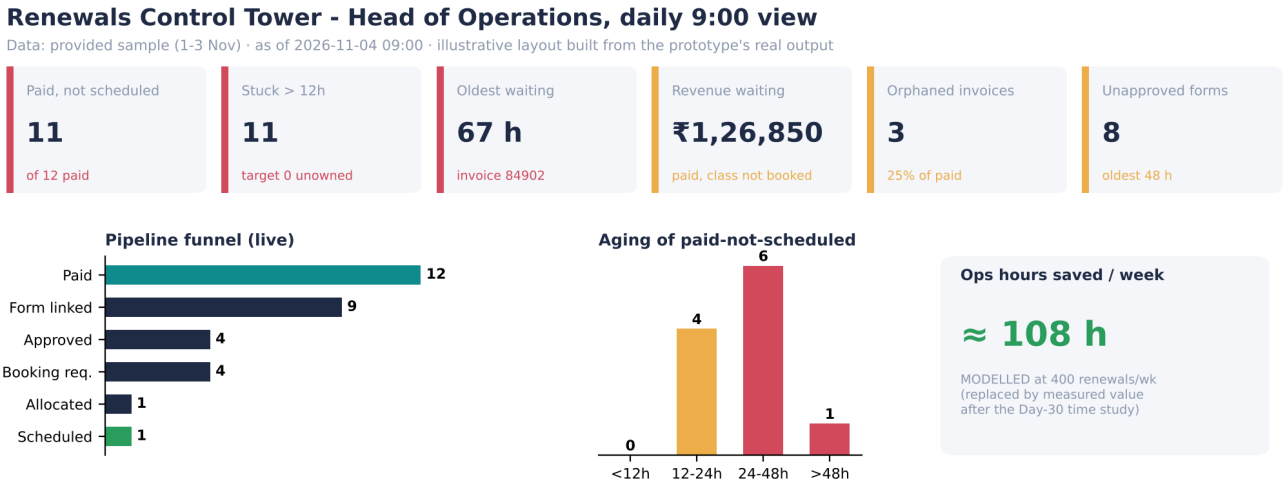
Things the engine caught that a human skim would likely miss: the Cyrillic homoglyph in 84913's name (a name-based lookup fails silently); that 84907's form amount equals *net* not gross; that 84908's duplicate was filed 5 minutes apart; that 84905's 'full' batch includes the renewing student; that the form for 84913 points to a non-existent invoice yet matches 84913 on user, student, educator and amount.

Known limits: *FALSE_BATCH_FULL assumes the active count already includes the renewing student (confidence marked MEDIUM, validate in shadow mode); no approval table exists, so every discount is flagged; the proposed first-session date is simplified; the engine outputs CSV/JSON rather than writing back to MySQL.*

2-3 minute Loom outline (optional deliverable): (1) 20 s: the 13 invoices and the funnel; (2) 60 s: run the script, open exceptions.csv and show 84911 and 84913; (3) 40 s: open messages.md, show the parent and Ops drafts; (4) 20 s: dashboard and rollout plan. Script in docs/loom_script.md.

Deliverable D - Dashboard, SOP and 30-60-90 plan

D1 · The Head of Operations' 9:00 dashboard



Exception queue - one owner per card, SLA timer running

Invoice	Prio	Reason code(s)	Owner	₹ paid	Hrs since pay	SLA	Next action (auto-drafted)
84907	P1	DISCOUNT_NO_APPROVAL_REF	Ops Lead / Finance	₹7,600	38 h	4 h	Approve/reject 20% discount
84911	P1	PAYMENT_NOT_CONFIRMED	Payments / Ops	₹12,000	20 h	4 h	Poll gateway; send fresh link
84909	P2	PARTIAL_PAYMENT + FORM_AMOUNT_MISMATCH	Ops Lead / Finance	₹5,250	23 h	4 h	Decide pro-rata vs hold; fix form
84912	P2	BATCH_CHANGE_REQUESTED	Ops Scheduling	₹9,000	17 h	6 h	Offer 2-3 slots to parent
84904	P3	PAYER_CONTACT_MISMATCH	Parent -> Ops	₹12,000	46 h	4 h	Confirm payer via registered no.
84906	P3	ORPHANED_INVOICE	Parent -> Ops	₹18,000	41 h	4 h	WhatsApp parent: same batch?
84910	P3	ORPHANED_INVOICE	Parent -> Ops	₹11,000	22 h	4 h	WhatsApp parent: same batch?

Refresh every 15 minutes from the event log (Metabase on the read replica, or Google Sheets for week 1). Red/amber thresholds: any P1 item or any unowned stuck invoice turns the header red. Click-through opens the exception card with the evidence. 'Hours saved' is a model until the time study replaces it with measured values.

D2 · SOP: who does what, when

Role	Daily responsibility	Rule of thumb
Head of Ops	9:00 five-minute scan of the six tiles; asks 'why is anything red'	Zero P1 left at 18:00; weekly review of SLA breaches
Ops lead	9:15 assigns P1/P2 cards; owns discounts, partial plans, price mismatches; 5% random audit of auto-approvals	Decide in the card: Approve / Reject / Call parent. Three clicks max
Ops agent (on shift)	Works payer confirmations, orphan escalations and batch-change slots; sweeps at 11:00, 15:00, 18:00	Never edit the booking tables by hand; use the card so the audit log stays complete
Weekend on-call agent	10:00-18:00 for P1/P2 only (weekend batches exist, e.g. 9205 and 9208)	Automated steps run 24x7; the SLA clock does too
Teacher	Sends renewals only through the link button; discounts only through a code; replies to nudges	Never accept payment screenshots; ask the parent to retry the link
Finance	Approves discount codes in advance; clears partial-payment decisions	Policy lives in a code table, not in chat

SOP for any exception card: (1) read the one-line evidence; (2) choose Approve / Reject / Reach out; (3) the engine updates the tables, messages the parties, closes the card and logs who decided. A card that is not touched by 50% of its SLA pings the owner; at 100% it pings the next level; at 200% it reaches the Head of Ops.

D3 · Change management: 200+ teachers, 10 Ops, no broken live schedules

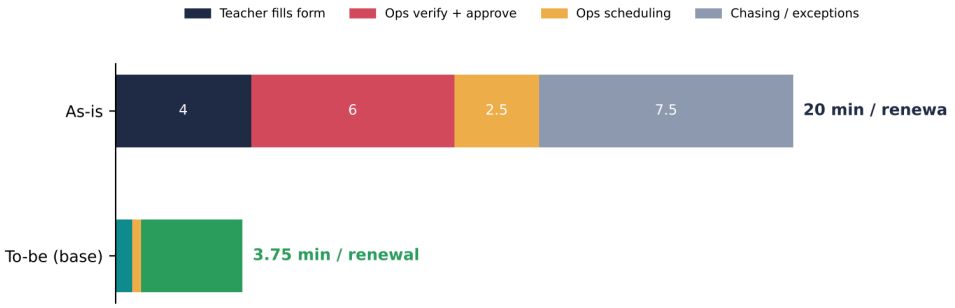
Wave	Scope	Entry gate	Exit gate / rollback trigger
0 Shadow (wk 1-4)	Engine runs on real data, writes to staging, customer-facing messages OFF. Ops keep working as today	Read access and data checks complete	≥ 95% agreement with Ops decisions on 2 weeks of data; every disagreement explained
1 Pilot (wk 5-8)	~20 teachers from 2 categories; same-batch, fully-paid renewals only; WhatsApp confirmations on	Wave 0 gate met; capacity-logic fix live	0 double bookings, wrong-schedule rate < 1%; else auto-pause and revert to manual for that cohort
2 Scale (wk 9-11)	All teachers, same-batch happy path; payer confirm, orphan and failed-payment flows	Wave 1 gate met; Ops trained on cards	Touch rate falling; P90 TAT ≤ 6 h
3 Complete (wk 11-13)	Link-with-intent for all; slot picker for batch change; legacy form retired	Teacher adoption > 90%	Orphan rate 0; P90 TAT ≤ 4 h

- **Protecting live classes:** the bot only inserts rows; schedule writes go through the same validation as Ops' today; nothing is auto-scheduled for a slot that starts within 2 hours without an Ops check; Ops can undo any bot booking within 24 h; a feature flag per teacher cohort and a global kill-switch.
- **Teachers (200+):** one 3-minute video and a one-page WhatsApp card in English and Hindi; 10 category champions who answer first-line questions; and a visible win: every confirmed renewal sends the teacher an instant 'credited to you' message. A teacher who sees it works stops skipping the step.
- **Ops (10):** the job shifts from data entry to exception ownership. Train on the card flow, name an owner per reason code, retire the approval sheet only after Wave 2, and share the hours-saved number so the change reads as capacity returned, not headcount at risk.
- **Parents:** they only see better service: instant confirmation, one-tap choices, and never a request to pay twice.

D4 · 30-60-90 day plan

	Days 1-30: see it and stop the bleeding	Days 31-60: automate the happy path	Days 61-90: scale and retire the form
Do	Baseline on 8-12 weeks of production data (replace this sample's rates); 2-day time-and-motion study; daily 'paid-not-scheduled' list and dashboard v1 in Sheets/Metabase; fix the capacity check; discount-code table and partial-payment policy; shadow-mode engine	Link-with-intent button; webhook → engine → booking/schedule for same-batch full-pay renewals in the pilot cohort; WhatsApp confirmations; exception cards live with SLA timers	All teachers on link flow; slot picker for batch change; retire the form and the approval sheet; monthly rule review; hand over a playbook to Ops
Measure	Baseline TAT, orphan rate, touch rate, class interruption days	Zero-touch ≥ 50%; P90 TAT ≤ 6 h; wrong-schedule rate < 1%	Zero-touch ≥ 65%; P90 TAT ≤ 4 h; orphan rate 0; ≥ 80 Ops h/week returned
Risk to watch	Data access delays; sample not representative	Teacher adoption; parents ignoring WhatsApp	Edge cases (siblings, refunds, time zones) outnumber the sample

D5 · Time and cost savings (model, assumptions explicit)



Assumptions (all to be replaced by the Day-30 time study): 400 renewals/week (the brief says 'hundreds'); minutes per renewal as shown: teacher form 4, Ops verify and approve 6, Ops scheduling 2.5 averaged across renewals, chasing 7.5 averaged; to-be uses the sample's mix (67% zero-touch, 8% soft verify at 3 min, 25% human decision at 12 min) plus 0.5 min for the teacher to add intent to the link. **Result: ≈ 20 → 3.75 min per renewal, ≈ 108 Ops-and-teacher hours saved per week** (≈ 99 h in a conservative case with 50% zero-touch), roughly 2.5-2.7 FTE-equivalent on a 10-person Ops team, before a 90-day ramp factor. Run cost is modest: n8n self-hosted or low-tier cloud, Metabase OSS, and about 3 WhatsApp template messages per exception (≈ 400 × 35% × 3 ≈ 420 messages/week at the assumed volume; price per conversation to be quoted by the WhatsApp provider). Not modelled but likely larger: retention from fewer class interruptions, and recovered revenue from discounts and partial payments being governed.

Open questions for the live case review & limitations

- Is the 'active_students_count' inclusive of the renewing student? My FALSE_BATCH_FULL fix depends on it.
- Is the full 24-session term scheduled at once, or rolling? (84901 shows 2 rows for 24 purchased sessions.)
- What is the discount-approval policy today, and who may authorise a retention discount? What is the partial-payment rule (pro-rata sessions or hold)?
- How are time zones for GCC/US batches stored? Are Stripe invoices for non-India parents genuinely in INR?
- How many renewals per week, and what share are same-batch? The 67% zero-touch figure rests on 12 invoices.
- What happens to a seat between term end and renewal; is it held, or released to a waitlist?

Limitations

Small, masked, 3-day sample; no weekend, night or refund behaviour observable; no real WhatsApp, Slack or MySQL connections were made, so the prototype produces drafts and CSVs, not live actions; the time-savings model rests on stated assumptions, not a measured study. Everything is built so these gaps close in the first 30 days.

Appendix · Repository contents

Path	Purpose
src/reconcile.py, messages.py, metrics.py	Engine, drafted follow-ups, SLA metrics
outputs/	auto_approved.csv, exceptions.csv, messages.md/.json, invoice_audit.csv, metrics.json, reason_codes.csv
prompts/	triage_prompt.md (remark reader), message_drafting_prompt.md
tests/test_golden.py	13 hand-verified decisions
docs/	n8n_workflow.md, loom_script.md, this report